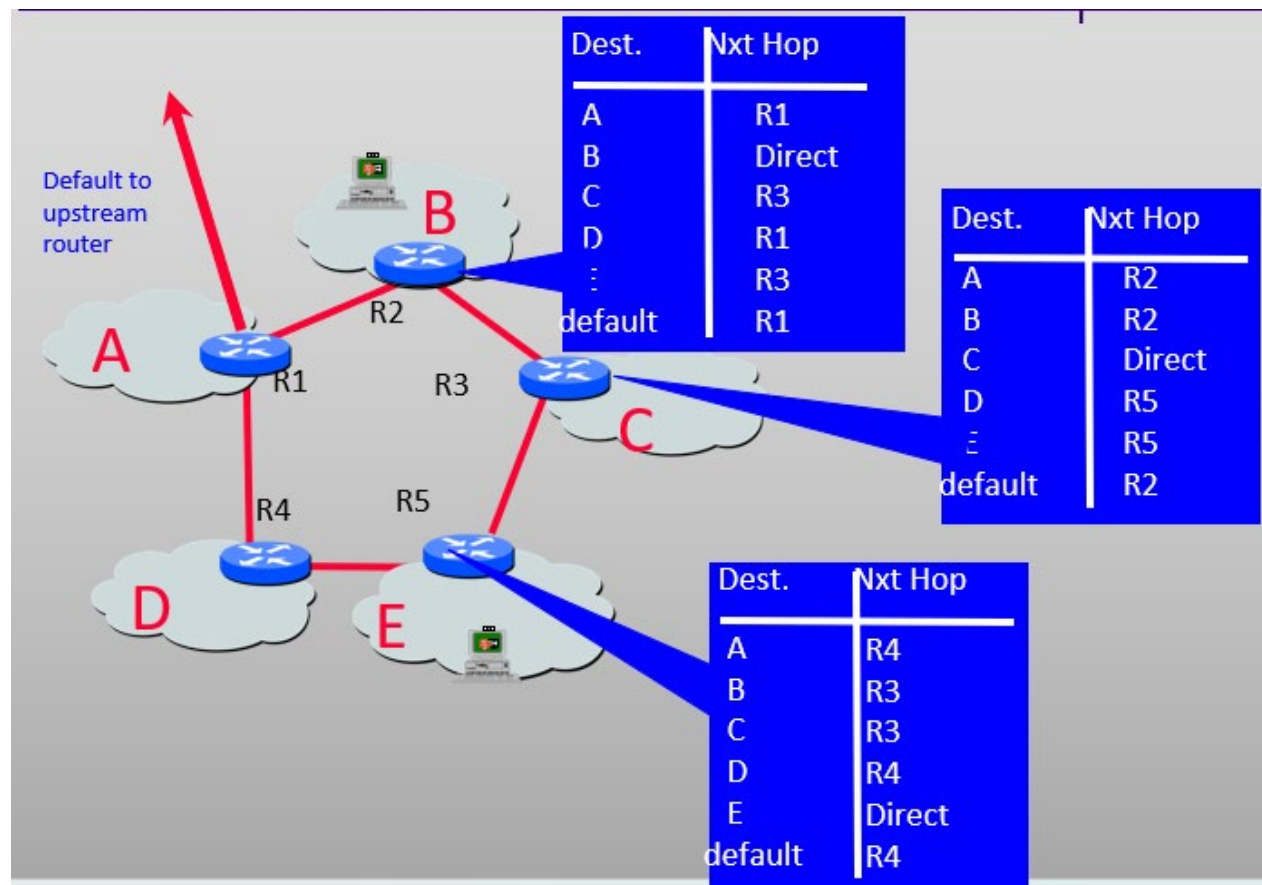


Core Network Requirements

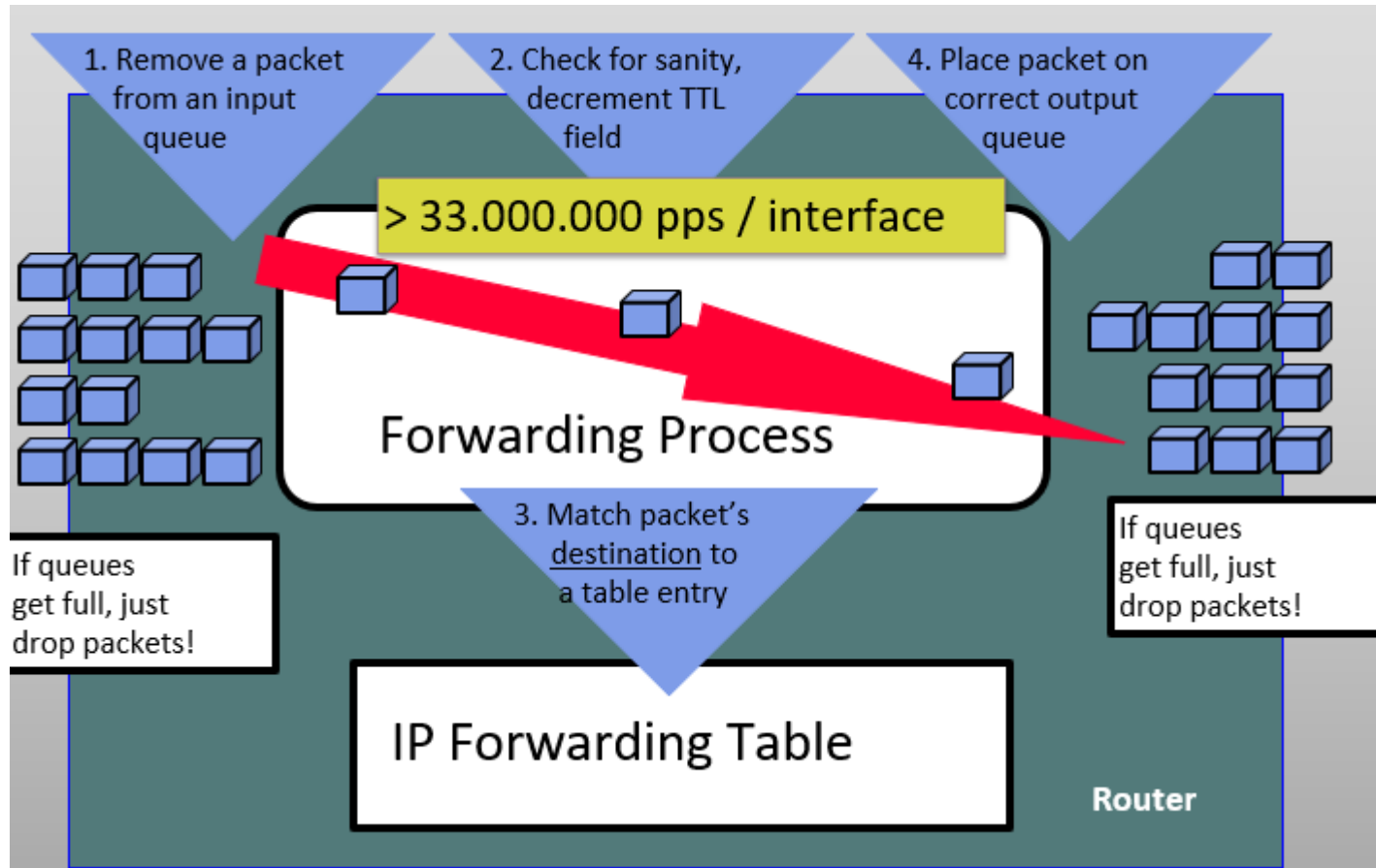
- High demands in terms of speed, bandwidth and load on existing infrastructure
 - Less than 30 ns to process 1 packet of 1500 Bytes at a rate of 400 Gbps
 - switch more than 33,000,000 packets per second (pps) per interface!
- User Requirements
 - Handle Quality of Service (QoS) and Classes of Service (CoS) for different customers

IP Forwarding

- Based on next-hop routing

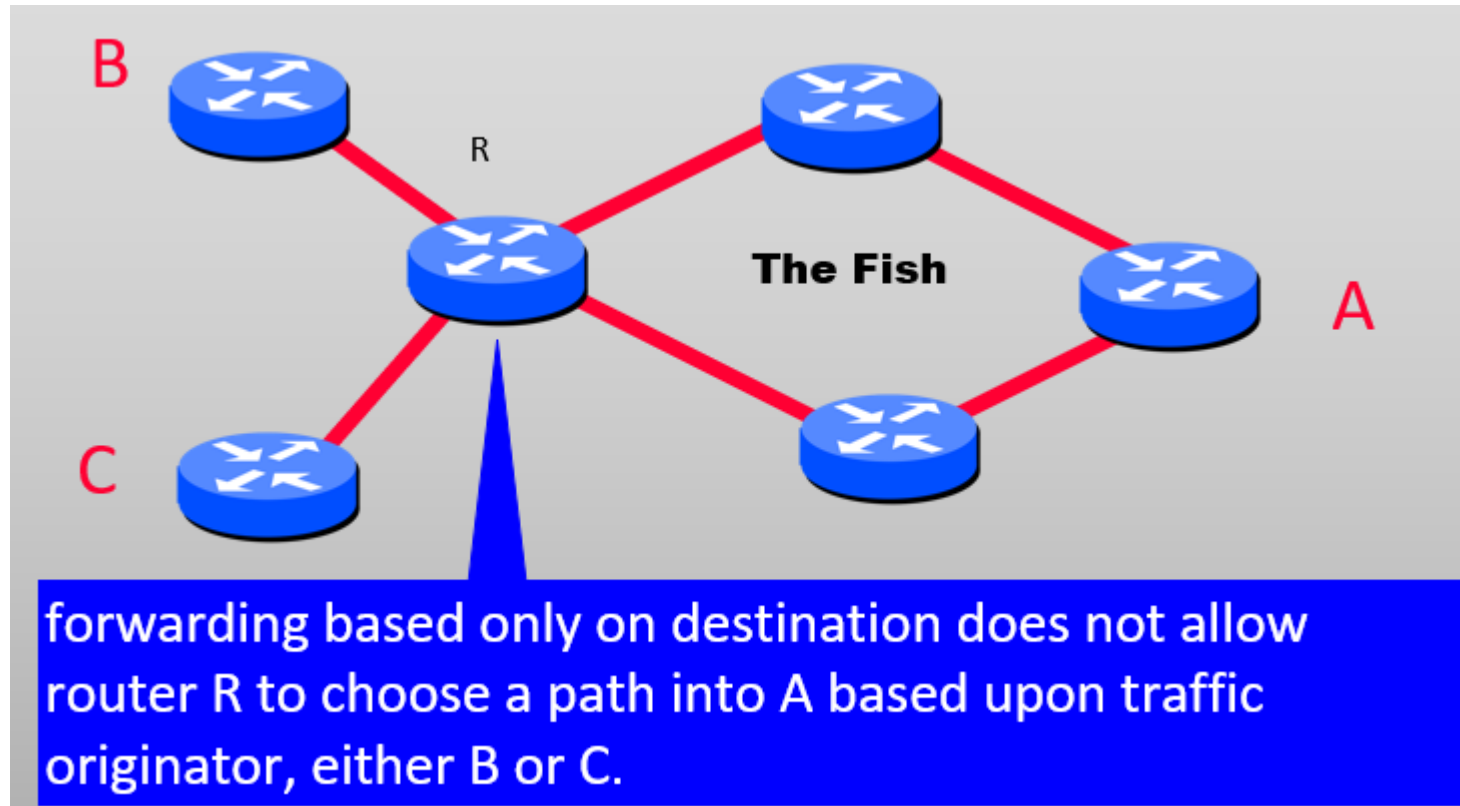


IP Forwarding Process



IP Forwarding

- Based on IP destination



- The traffic originator is the customer

Multiprotocol label switching (MPLS)

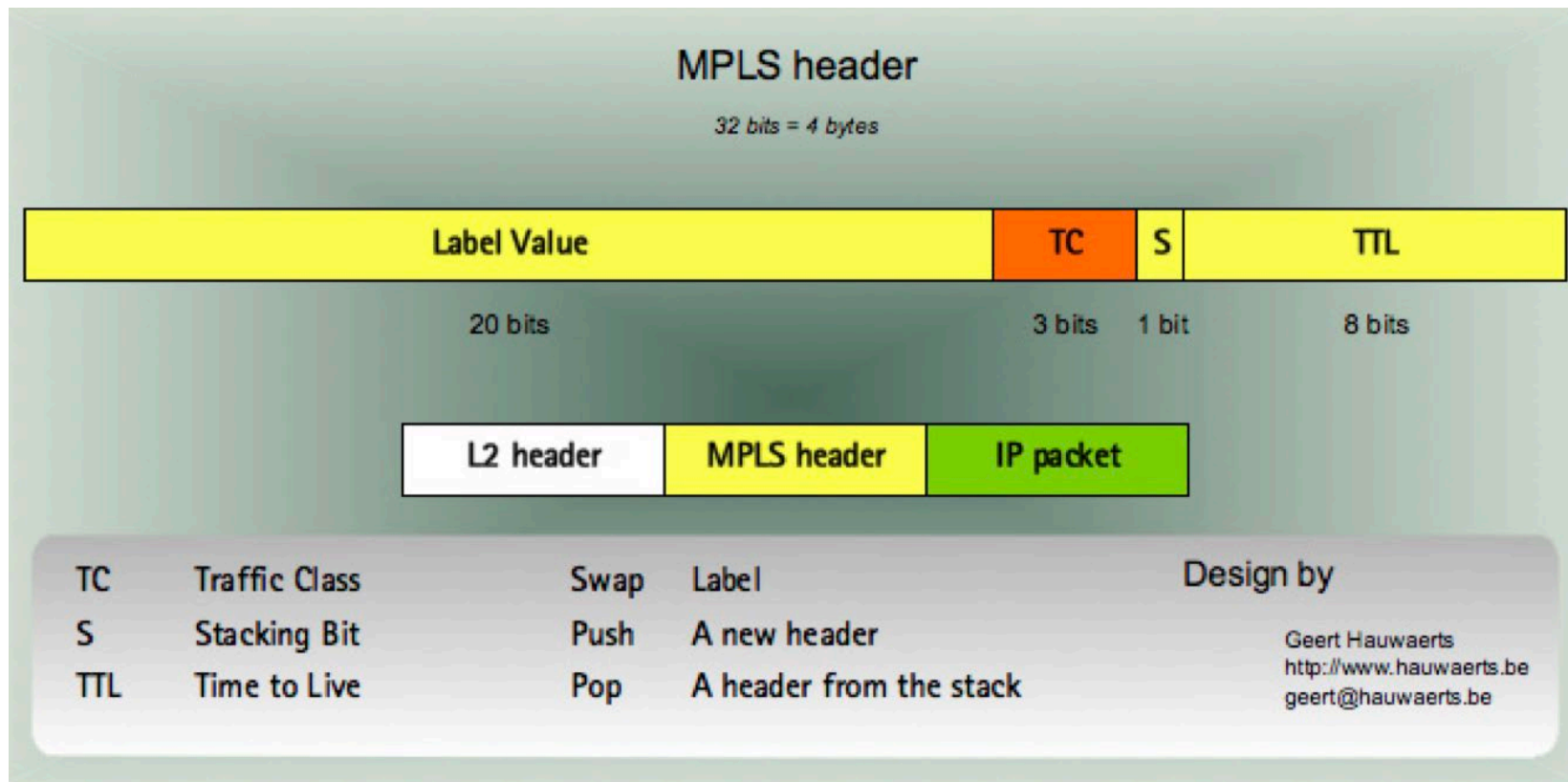
- What is it?
 - Labeling packets with fixed-size labels at the network entrance
 - Tag-based routing and tag exchange within the network
- Why?
 - Ultra-Fast “Forwarding”
 - A middle term between packet and circuit switching
 - It was born to coexist better with ATM and Frame relay... But now it really replaces it... (and we can have ATM circuits over MPLS)
- IP traffic engineering: manipulation of traffic flows, routing subject to restrictions, etc.
- Virtual Private Networks (VPNs):
 - L3vpn; l2vnp; Virtual Private LAN Service (vpls); Pseudo-Wire Emulation (pwe3)
- QoS Mechanisms
 - DiffServ-TE, voice and video over IP

MPLS Past and Current Scenario

- Organizations where performance, security, and reliability are critical operations rely on MPLS for its operation
- MPLS was introduced by IETF (Internet Engineering Task Force) in the late 1990s as a solution to overcome the limitations of both IP and ATM
- MPLS was standardized in the early 2000s, and telecommunications companies and internet service providers (ISPs) adopted the technology
- Throughout the 2000s, MPLS gained wide adoption across enterprises and ISPs.
 - MPLS-based VPNs became the preferred solution for connecting geographically dispersed offices or data centers securely.
 - MPLS became the backbone of many large-scale IP networks
- IP MPLS VPN Service Market is projected to grow from USD 68.18 Billion in 2025 to USD 115.40 Billion by 2034, with an annual growth rate (CAGR) of 6.20% during the forecast period (2025 - 2034)¹

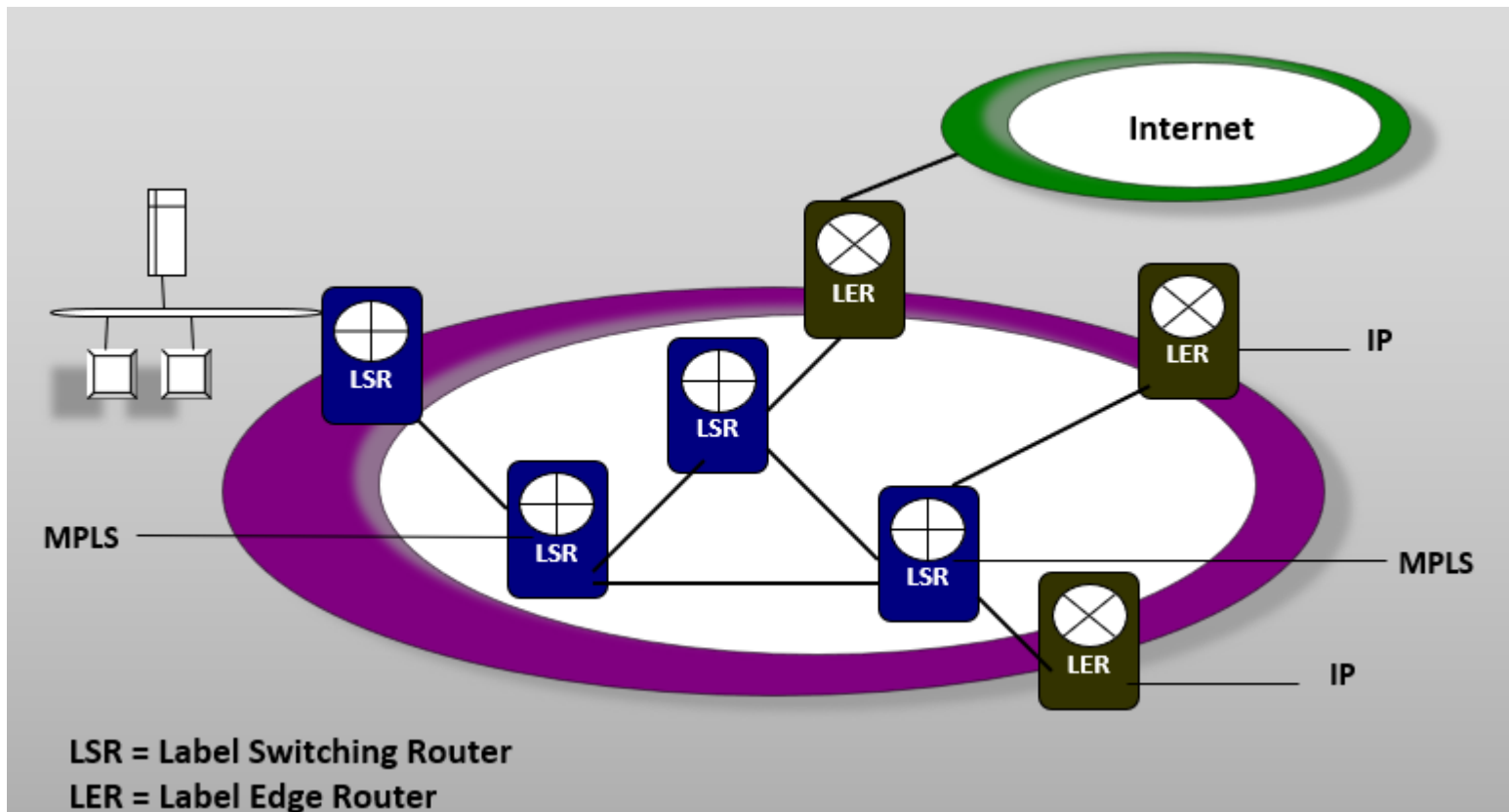
[¹Ip Mpls Vpn Service Market Size, Share Growth | Outlook 2034](#)

MPLS - Tags

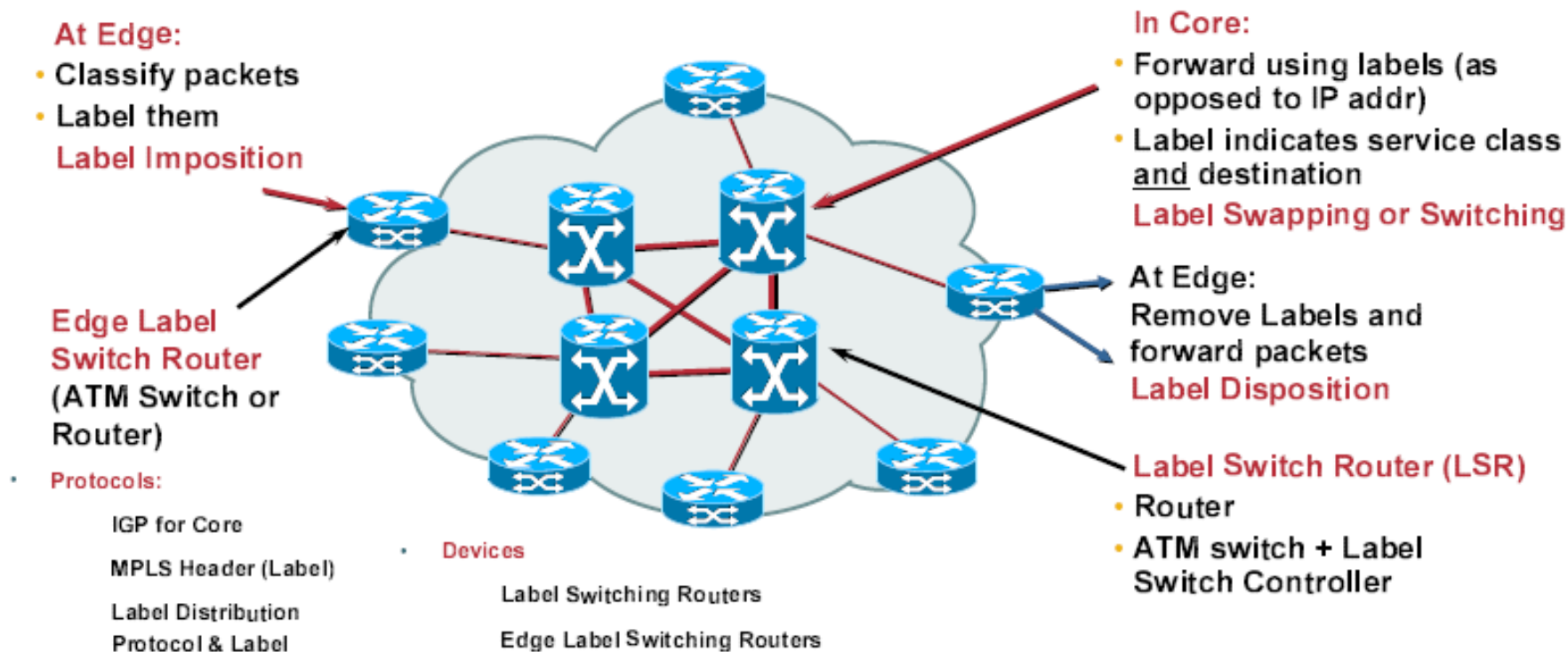


- Originally the 3 bits in red were called EXP (experimental). Later, RFC 5462 changed the name of the “EXP field” to “Traffic Class field” (“TC field”)

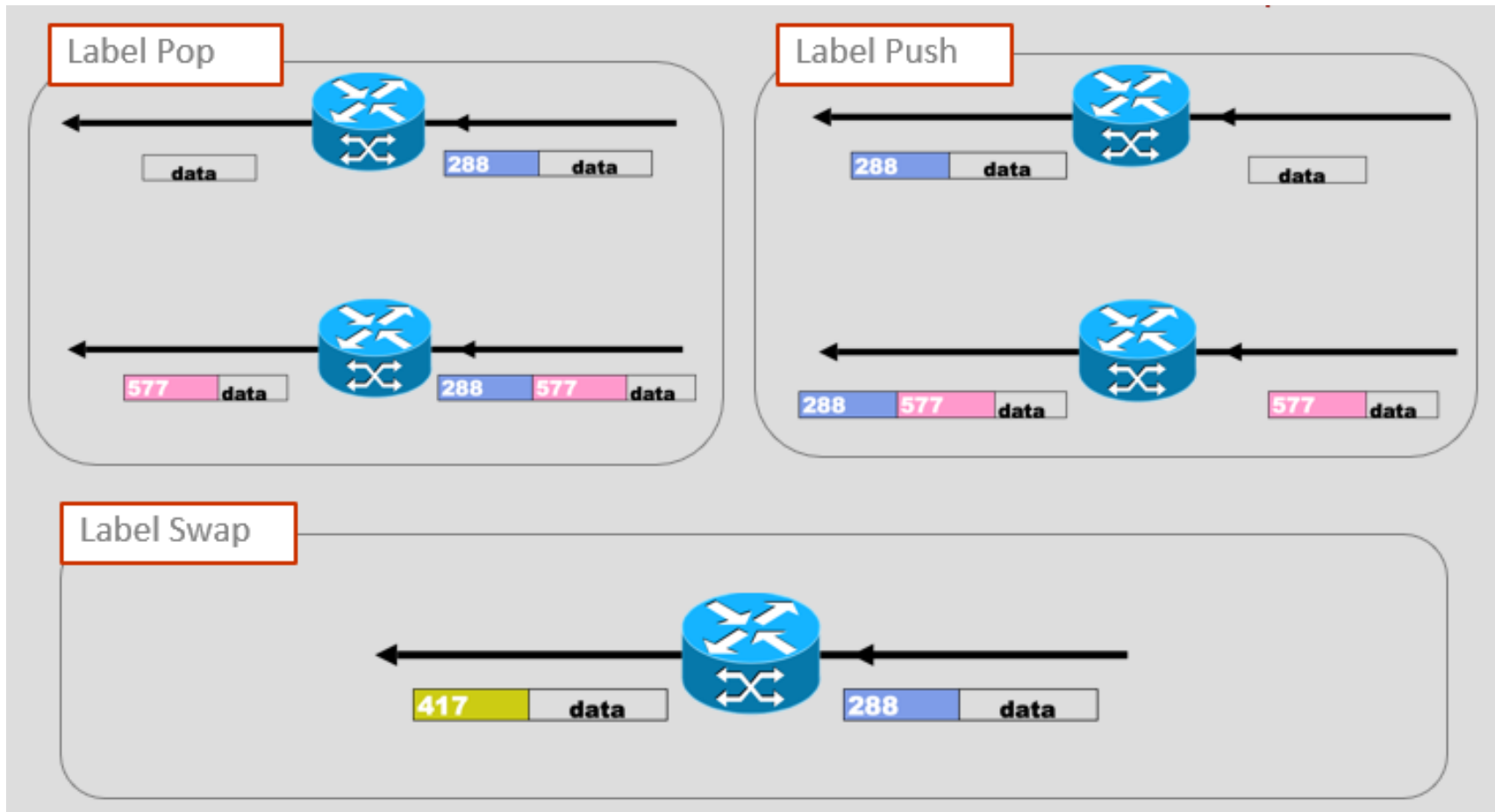
MPLS - Network Model



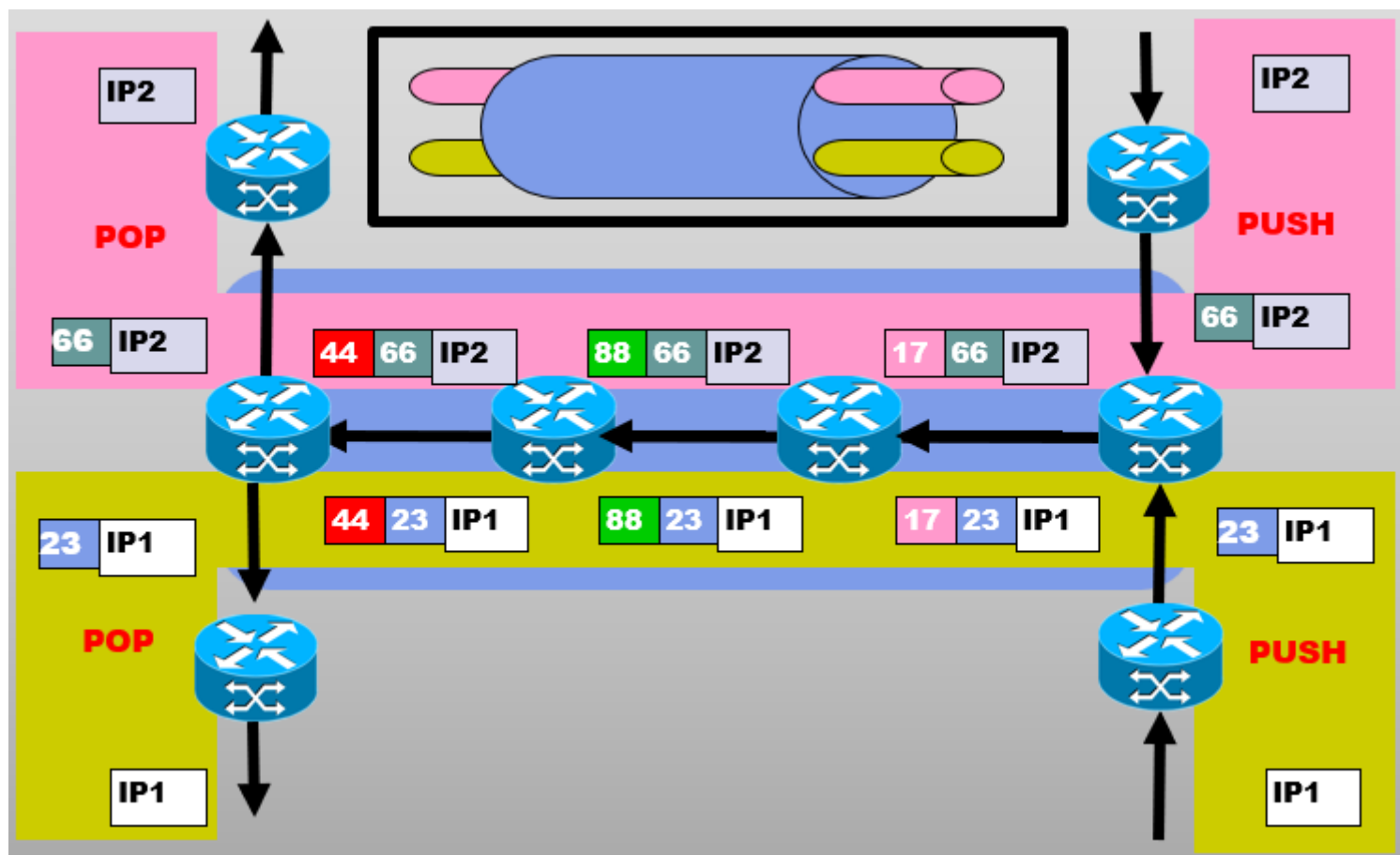
MPLS Components



MPLS – Label Push, Pop, Swap



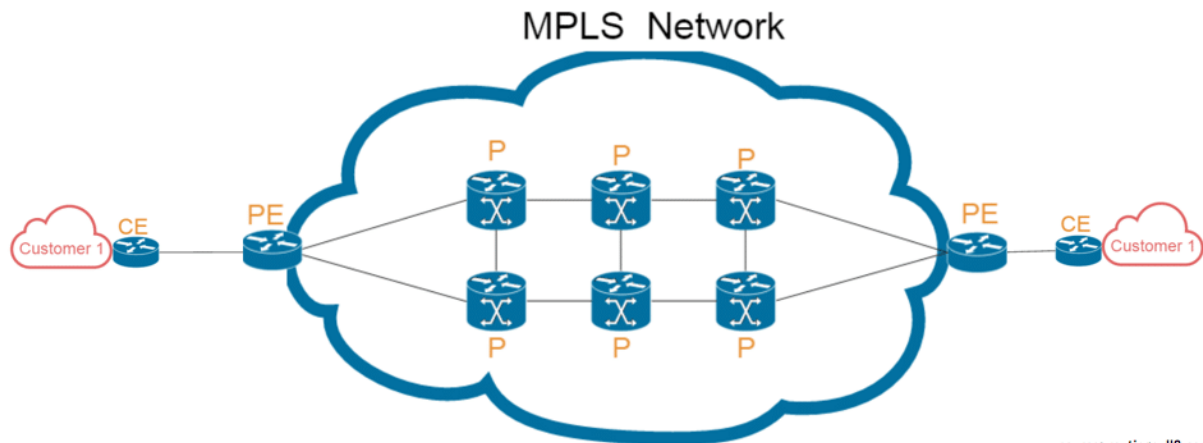
LSP Hierarchy - Tag Stack



MPLS Concepts

- LSR: Label Switching Router
- LER: Label Edge Router
 - Useful term not in standards
- LDP: Label Distribution Protocol
- LSP: Label Switched Path
- FEC: Forwarding Equivalence Class
- FIB: Forwarding Information Base
- LIB: Label Information Base
- Label Stack
 - Label Imposition (push), Label disposition (pop) e Label swapping

Typical Network Core Using MPLS



- **Provider Router (P)**
 - Element of the core network of the service provider
 - Forward data based on labels
 - Focus on label switching for traffic within the MPLS network
- **Provider Edge Router (PE)**
 - Located at the edge of the provider's network and interfaces with customer networks
 - PE routers maintain routes to customer network
 - In a VPN setup, PE routers handle the distribution of customer VPN routing information and MPLS label distribution to support data in tunnels
- **Customer Edge Router (CE)**
 - Located in the customer's side of the network
 - Manage the customer network's internal routing and connectivity to the provider network
 - Do not have any knowledge of MPLS or the labels used in the provider's network

MPLS – Operation Mode

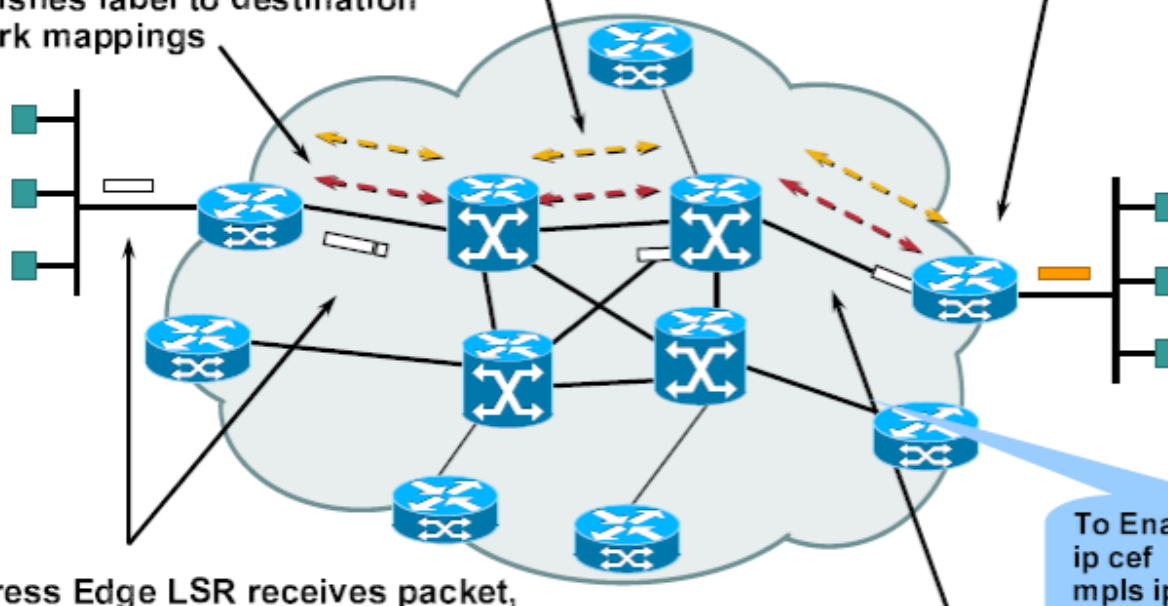
1a. Existing routing protocols (ie: OSPF, IS-IS, EIGRP) establish reachability to destination networks

1b. Label Distribution Protocol (LDP) establishes label to destination network mappings

4. Edge LSR at egress removes label and delivers packet

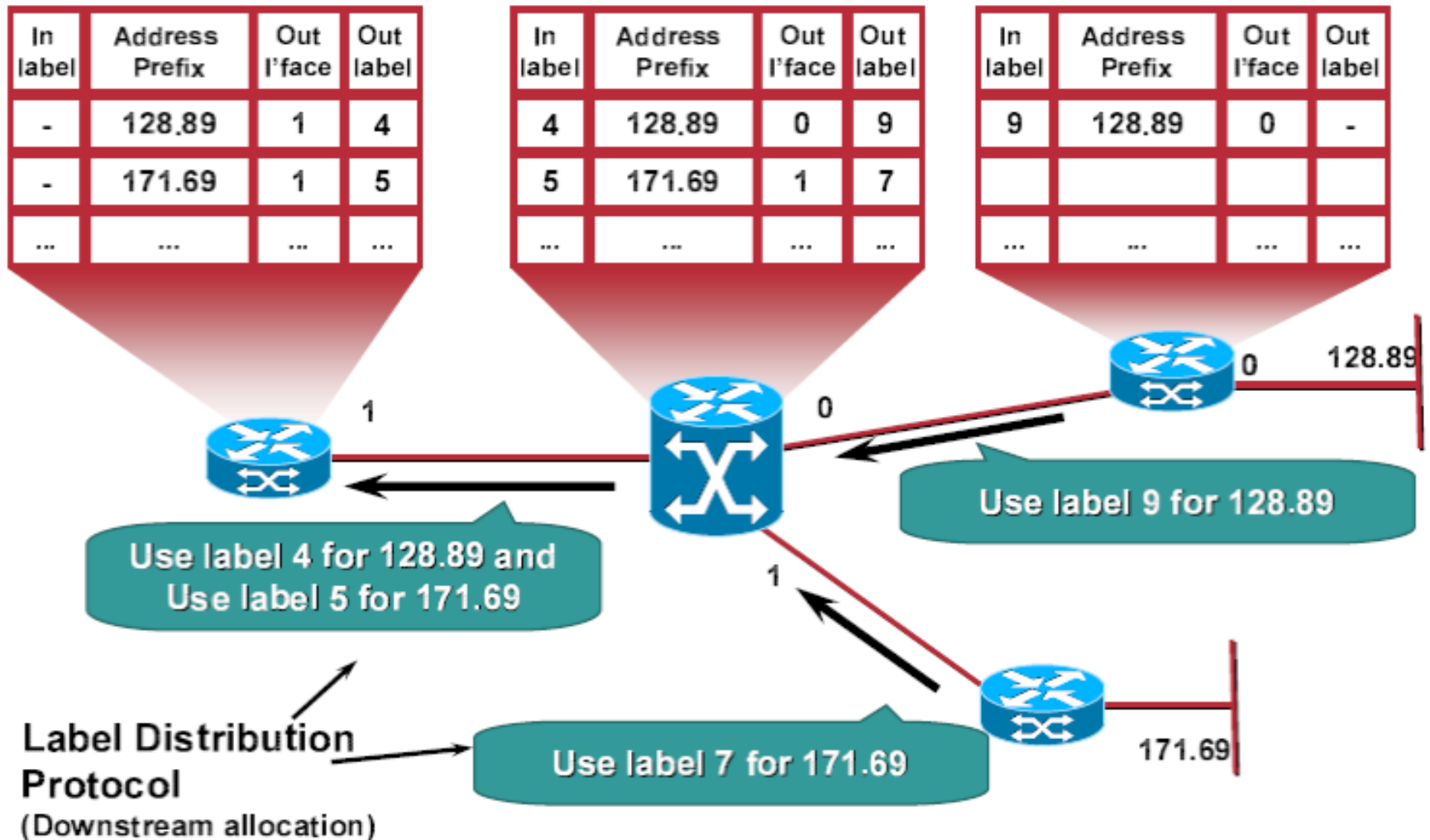
2. Ingress Edge LSR receives packet, performs Layer 3 value-added services, and “labels” packets

3. LSR switches packets using label swapping



To Enable mpls:
ip cef
mpls ip
mpls label protocol ldp

MPLS – Label Distribution

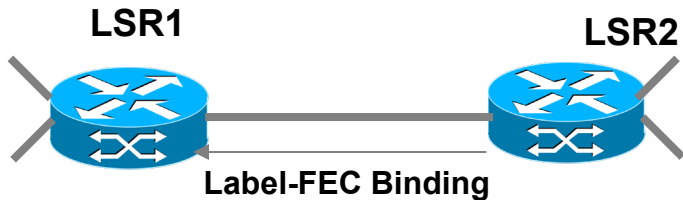


Forwarding Equivalence Class (FEC)

- FEC
 - Group of packets that share the same transport requirements
- In a FEC group, all data packets receive the same treatment
- MPLS FEC is established at the network input
- FEC is based on service requirements, related to a set of packets, or related to a prefix
- Each router builds its own forwarding table (Label Information Base) specifying how to forward data
 - Label Information Base, LIB, relates FECs and Labels

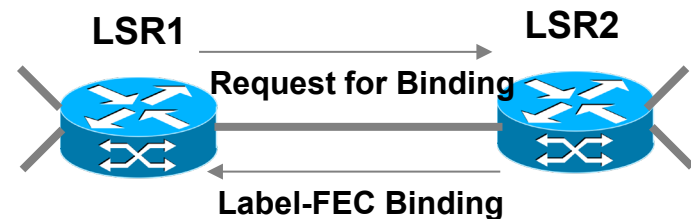
MPLS – Label Distribution

Downstream Label Distribution



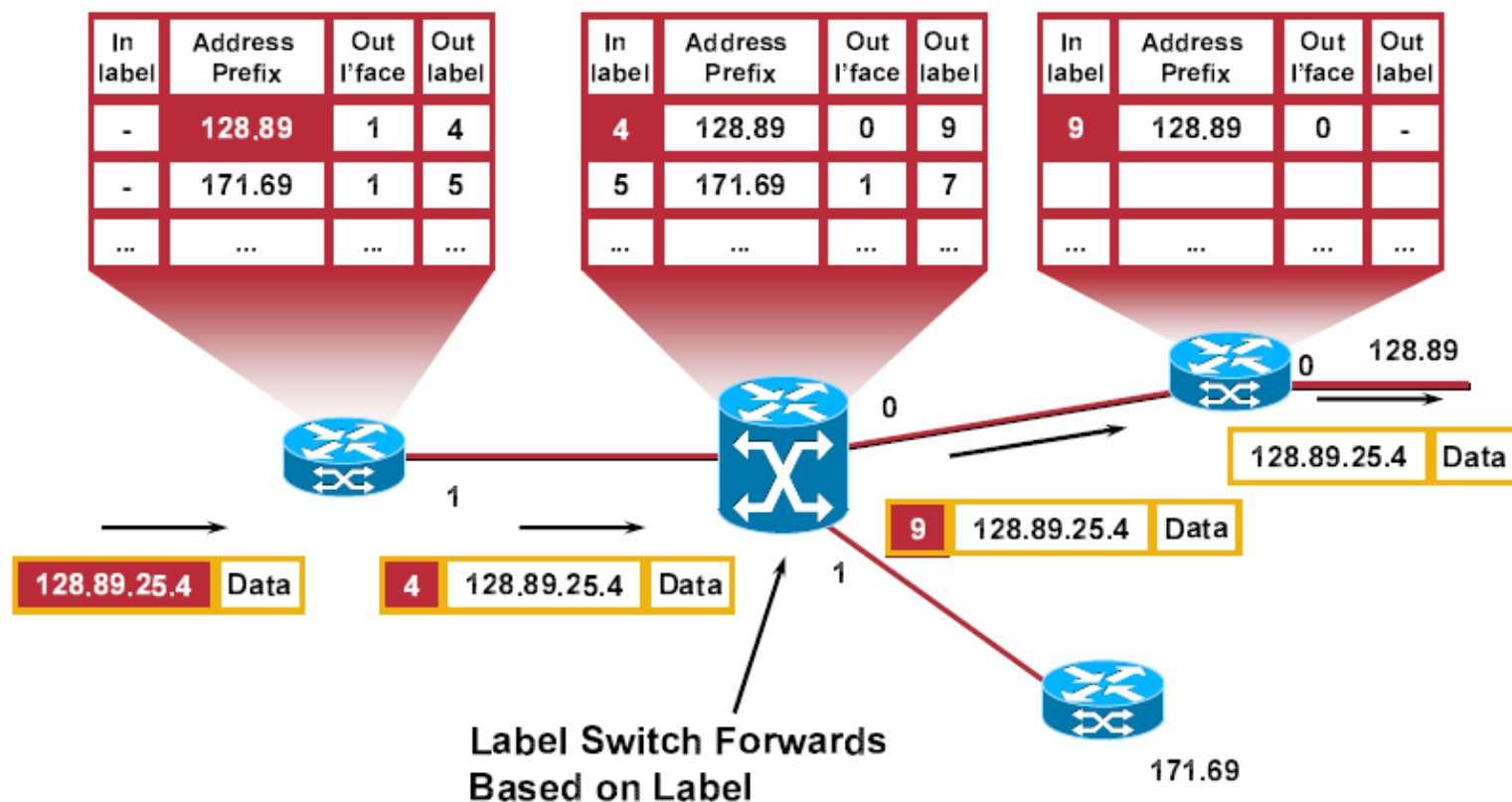
- LSR2 and LSR1 are LDP adjacent; LSR2 is the downstream LSR;
- LSR2 discovers a 'next hop' for an FEC sending class
- LSR2 generates a tag for this FEC and announces this association to LSR1
- LSR1 inserts the FEC-label association into its forwarding tables

Downstream-on-Demand Label Distribution



- LSR1 recognizes LSR2 as its next hop for a given FEC class
- LSR1 makes a tag request for this FEC to LSR2
- If LSR2 recognizes the FEC and has a next hop for it, it creates an association and responds to LSR1

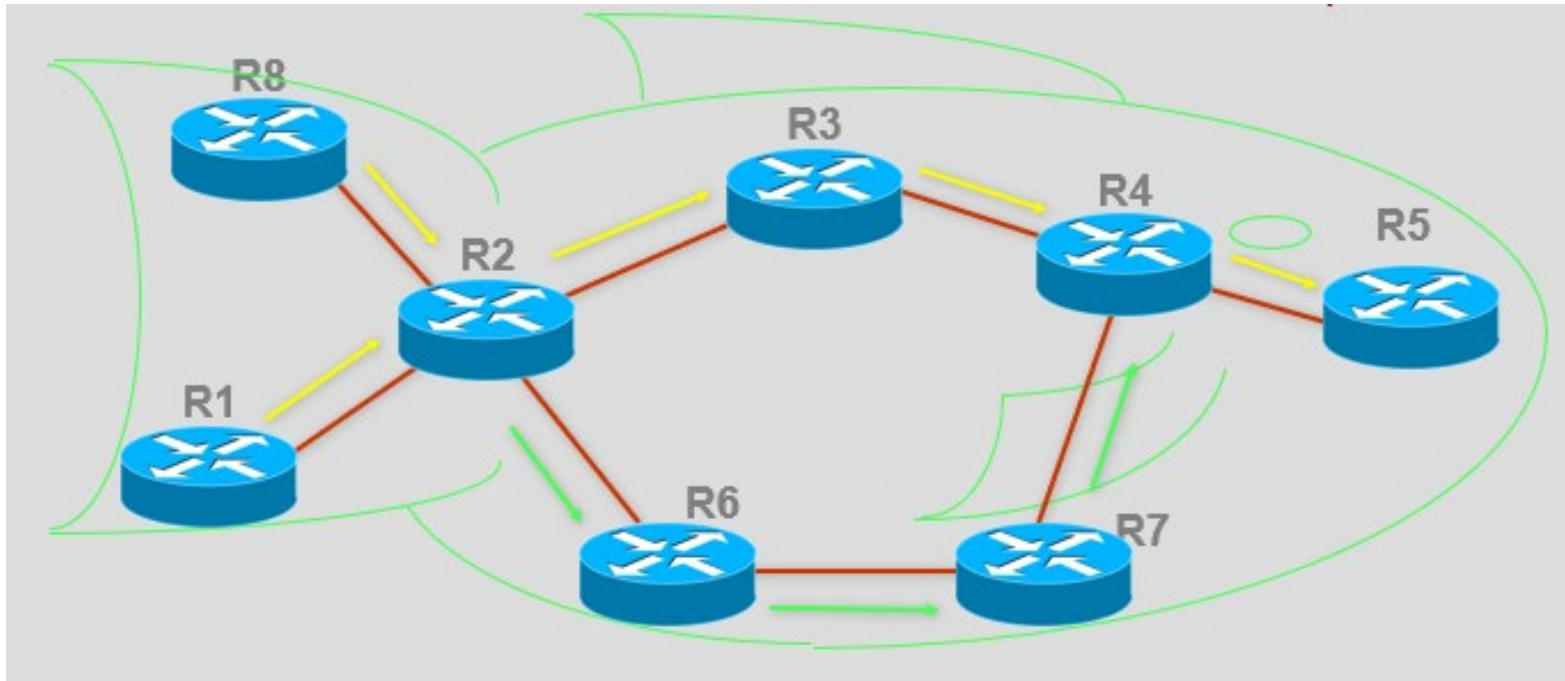
MPLS – Packet Forwarding



MPLS – Label Distribution

- LDP Protocol
 - Protocol to exchange Label Mapping Information
 - Extensible: Uses a TLV (Type-Length-Value) format to allow extensions
- It does not intend to be a routing protocol and depends on an IGP protocol that calculates the best routes
- Allows you to automate, with little or no configuration, the distribution of labels:
 - Neighbor discovery with Hello, using UDP port 646 and multicast
- Maintains active LDP session with neighbors using a TCP connection on port 646
 - Exchange of Label Mapping messages from Label withdraw
 - Distribution in two modes: Ordered (only after receiving a valid label from the next router does it generate its own) and Independent
 - Conservative retention (only keep the tag if it came from the next hop) or liberal (keep them all, even if they are not from the next hop)

The “Fish”



Normal IP routing takes traffic via the shortest path!



Alternative path exists but is always low in use!

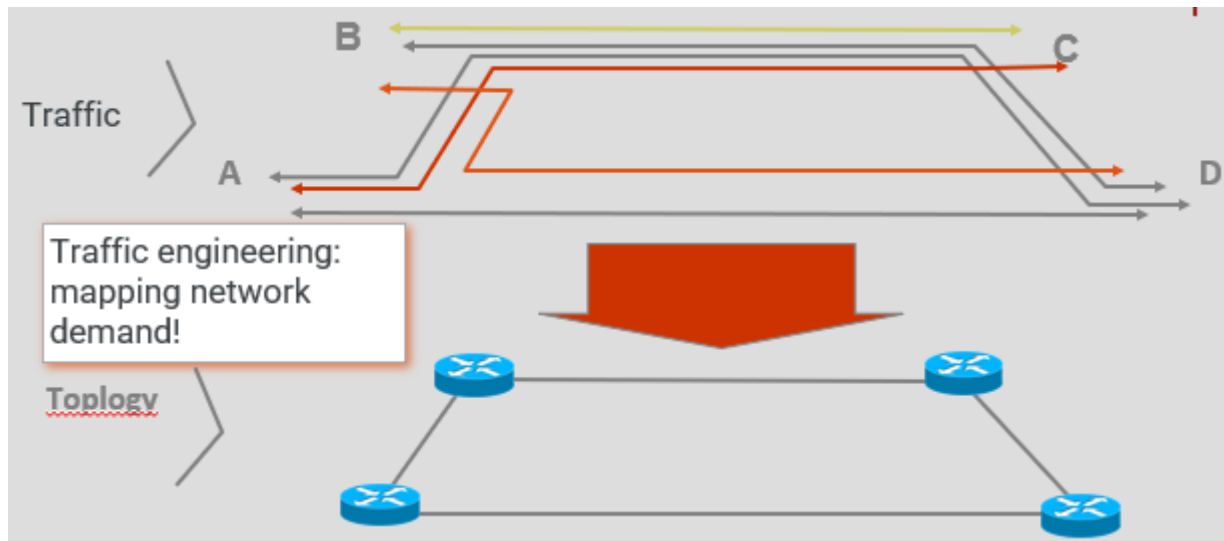
Core and Access Networks (Redes de Acesso e Núcleo)

Traffic Engineering

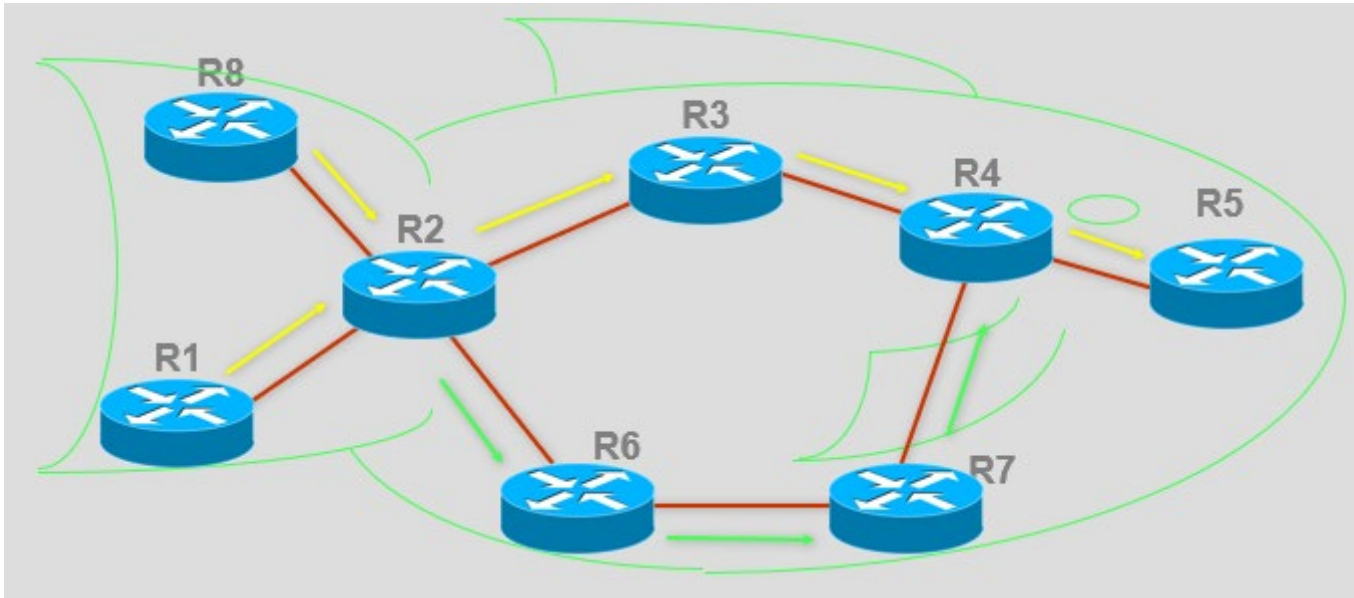
- It has nothing to do with QoS
 - but can improve QoS
- It's about reducing costs by making better use of existing resources!
 - Map demand with availability
- Take advantage of the near “connection-oriented” features of MPLS
 - Distribute traffic based on bandwidth demand or availability

Traffic Engineering

- Maximize the utilization of links and nodes across the entire network topology
- Call engineering to achieve desired delay times and service levels
- Spread traffic across network links, minimize the impact of an isolated failure
- Ensure free capacity to re-route traffic in case of failure
- Respect the policies imposed by the operator



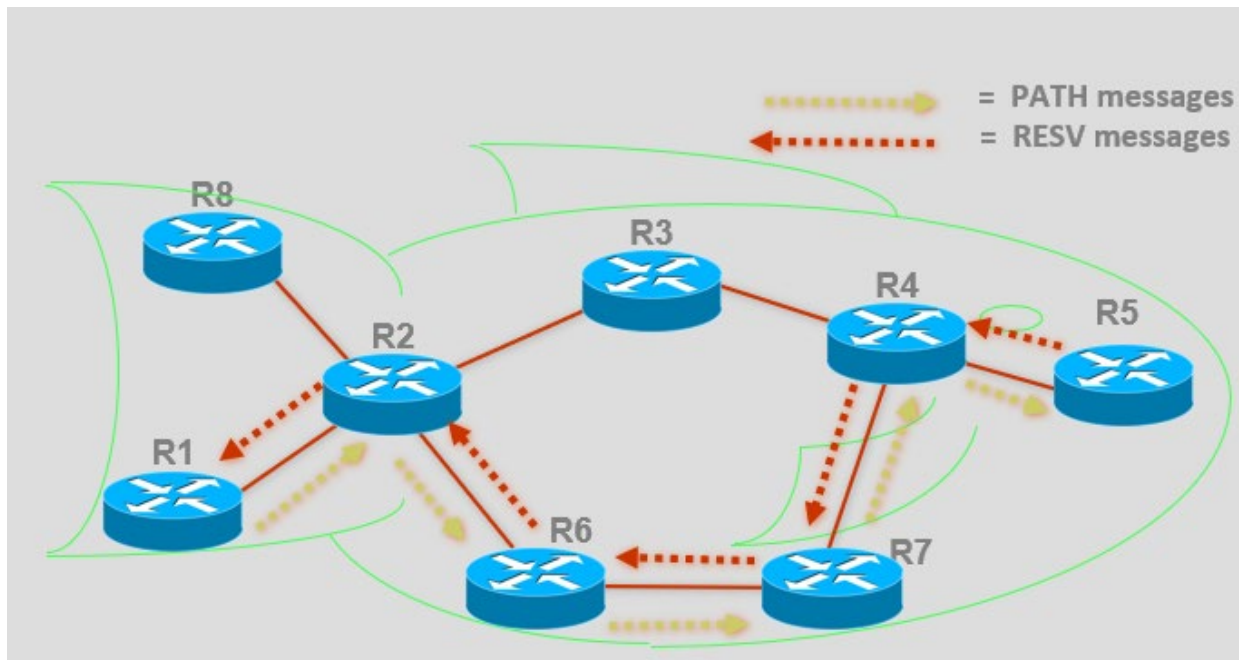
MPLS-TE: A simple example



- What can we do, with MPLS, to not always use the same path?

MPLS: RSVP Path Setup

- How to distribute labels on routes that are not optimal routes?
 - RSVP signaling and resource reservation protocol
 - PATH message: [R1-R2-R6-R7-R4-R5] “Can I have 40Mb along this path?”
 - RESV Message: [R5-R4-R7-R6-R2-R1] “Yes, and here’s the label to use”
 - LFIB filled along the route



MPLS: RSVP Path Setup

- Establishment of routes with the Resource Reservation Protocol (RSVP)
- Ingress LER node sends RSVP Path message towards egress LER node
- Path message has an active “Router Alert” option, which requires all routers on the route to inspect it and make the necessary modifications
- The Path message contains the following objects:
 - Label Request Object (LRO) requesting labels for the route
 - Explicit Route Object (ERO) list of route addresses to use
 - Record Route Object (RRO) requests memory of the route (all routers that process the message add their address to the message)
 - TSpec the bandwidth reservation requested by the source

MPLS: RSVP Path Setup

- Establishment of routes with RSVP
- Egress LER node responds with Resv message
 - The Resv message travels exactly the opposite route of the Path message: each router knows where it received it and where it should send it
 - The Resv message contains the following objects:
 - Label Object (LO) the requested MPLS label (only has local meaning, on the link between the two neighboring routers)
 - Record Route Object (RRO) requests memory of the route, as in the Path message (allows cycles to be detected)
 - RSVP requires periodic exchange of messages to “refresh” status

MPLS: RSVP Path Setup

- Ways to establish LSP paths:
 - LDP distributes the labels that MPLS uses, when the path is determined by the IGP protocol;
 - Labels distributed based on the best IGP paths!
 - RSVP is used to establish paths that are not determined by the best IGP path (**paths other than the shortest**)
 - Only the node at the end of the route can change it
 - Link state protocol (IS-IS or OSPF) required to announce bandwidth availability and calculate paths with bandwidth restrictions
 - RSVP signaling can cross domains
 - BGP

MPLS-TE: A simple example

- Activating fast forwarding (global command)

```
Router(config) #ip cef distributed
```

- Enable label distribution protocol (global)

```
Router(config) # mpls ip
```

```
Router(config) # mpls label protocol ldp
```

```
Router(config) # mpls ldp router-id loopback 0
```

- For each interface

```
Router(config) # interface ethernet 0
```

```
Router(config-if) # mpls ip
```

```
Router(config-if) # mpls mtu 1504
```

MPLS-TE: A simple example

- Check configuration

```
show ip cef summary
```

```
show interfaces
```

```
show mpls ldp discovery
```

```
show mpls ldp neighbor
```

```
show mpls interfaces [detail]
```

```
show mpls forwarding a.b.c.d detail
```

MPLS-TE: A simple example

- Enable traffic engineering on all routers:

```
interface loopback 0
```

```
ip address 192.168.1.1 255.255.255.255
```

- In general, and in all interfaces that may be part of a tunnel:

```
Router(config)# mpls traffic-eng tunnels
```

```
Router(config-if)# mpls traffic-eng tunnels
```

```
Router# show mpls traffic-eng summary
```

```
Router# show mpls interfaces
```

MPLS-TE: A simple example

- The most attractive capability of this engineering is the possibility of reserving bandwidth
- It is necessary to say, interface by interface, of those that may be used in the tunnel, what bandwidth can be reserved.

```
Router(config-if) # ip rsvp bandwidth <total-  
reservable> <maximum-per-flow>
```

...

```
Router# show ip rsvp interface
```

MPLS-TE: A simple example

- The path the tunnel follows is dynamically discovered according to specified bandwidth requirements

```
interface Tunnel0
```

```
ip unnumbered Loopback0
```

```
tunnel mode mpls traffic-eng
```

```
tunnel destination 192.168.1.254
```

```
tunnel mpls traffic-eng path-option 10 dynamic
```

```
tunnel mpls traffic-eng bandwidth kbps
```

MPLS-TE: A simple example

- But we can go a little further....

```
interface Tunnel0
```

```
ip unnumbered Loopback0
```

```
tunnel mode mpls traffic-eng
```

```
tunnel destination 192.168.1.254
```

```
tunnel mpls traffic-eng bandwidth kbps
```

```
tunnel mpls traffic-eng path-option 10 explicit path1
```

```
tunnel mpls traffic-eng path-option 30 explicit path2
```

```
tunnel mpls traffic-eng path-option 50 dynamic
```

- Manually giving two possible paths, trying in order, and trying dynamic construction to the end.

MPLS-TE: A simple example

- Explicitly saying where we want (or don't want) the paths:

```
ip explicit path path1
next-address 192.168.1.2
next-address 192.168.1.3
next-address 192.168.1.4
next-address 192.168.1.5
```

```
ip explicit path path2
  exclude-address 192.168.1.3
```

MPLS-TE: A simple example

- Then just route the traffic through the created tunnel
- Using for example static routes:

```
ip route 10.0.0.0 255.0.0.0 Tunnel0
```

...

```
ip route 192.168.1.1 255.255.255.255 Tunnel0
```

```
ip route 10.0.0.0 255.0.0.0 192.168.1.1
```

MPLS-TE: A simple example

- Or policy-determined routing:

Ethernet interface 0

```
ip policy route-map map-voice-in-tunnel
```

```
route-map map-voice-in-tunnel
```

```
match ip address 101
```

```
set interface Tunnel0
```

```
access-list 101 permit ip any 5.5.5.5
```

- Policy Based Routing (PBR) is used on the ingress interface (as soon as the traffic arrives and before forwarding decisions). Filtering can be sophisticated!

MPLS-TE: A simple example

tunnel mpls traffic-eng autoroute announce

- Or

tunnel mpls traffic-eng forwarding-adjacency

- Using the “autoroute announce” command
 - tells the router at the head to treat the tunnel as a link directly connected to the tail (no need to use static routes from the source to the tail of the tunnel)
- Using the “forwarding-adjacency” command
 - tells the router to advertise the tunnel in IGP routing protocols (to everyone)
 - The tunnel must be bidirectional
- You cannot activate IGP routing protocols in a tunnel, because:
 - They are unidirectional and normally the tunnels are internal to an area; start and end within the area; There is already a flood of LSAs or equivalent, so it is not necessary...

MPLS-TE: A simple example

- Enable IGP routing protocols
 - Mandatory “connection status”
 - In other words: OSPF or IS-IS

OSPF:

```
router ospf 20
```

```
mpls traffic-eng area X
```

```
mpls traffic-eng router-id LoopbackN
```

```
... (must have mask 255.255.255.255)
```